Movie Rental Analysis – MECE Framework

Capstone Project | Self Analysis | Matrix layout for Power BI report

1. Problem Statement

Build a Power BI dashboard for a DVD rental store (Sakila database, 16 tables) that helps the owner understand customer behavior, film inventory, staff performance and store operations. The findings feed simple recommendations for marketing campaigns, the film collection and staff training.

2. MECE Matrix Overview

Root node: **Improve Rental Store Business Performance (Sakila dataset, 16 tables)**. It splits into 6 analysis areas, each with 3 sub-analyses, in the same matrix style as the Pubs example.

Area	Sub-analysis 1	Sub-analysis 2	Sub-analysis 3
Customers Analysis	Country	Spending	New vs existing
Sales Analysis	Total revenue	Monthly trend	Average payment
Films Analysis	Top films	Category	Rate and rating
Inventory Analysis	Copies per film	Films with no copy	Returns
Staff Analysis	Rentals handled	Revenue	Average payment
Stores Analysis	Store revenue	Busy days / hours	Customers per store

3. Detailed Matrix by Area

Each block gives the sub-analysis, what to analyze, the key finding and the Power BI visual to plot.

3.1 Customers Analysis

Tables: customer, address, city, country, payment

Country	Spending	New vs existing
Analyze where customers come from.	Evaluate how much each customer spends.	Compare customers who joined after May 2005 with older ones.
Finding: 599 customers from 108 countries, 584 active (about 97%). Top: India 60, China 53, USA 36, Japan 31, Mexico 30.	Finding: average spend $112.53 (min $50.85, max $221.55). Top 10% of customers bring only 14.3% of revenue. Top spender: Karl Seal, $221.55.	Finding: 79 newer customers rent slightly less (25.4 vs 27.0 rentals) but started later.
Visual: filled map and bar by country.	Visual: top 10 customers bar, spend distribution.	Visual: clustered column, new vs existing.

3.2 Sales Analysis

Tables: payment

Total revenue	Monthly trend	Average payment
Measure overall revenue and payment count.	Analyze sales patterns month by month.	Evaluate the typical payment size.
Finding: $67,406.56 from 16,044 payments.	Finding: $4,823 (May 2005), $9,630 (June), peak $28,369 (July), $24,070 (August). May, June and February have less data.	Finding: $4.20 per payment.
Visual: KPI cards.	Visual: line or column chart by month.	Visual: KPI card with monthly trend line.

3.3 Films Analysis

Tables: film, category, film_category, rental

Top films	Category	Rate and rating
Identify most rented films.	Compare revenue by film category.	Evaluate rental rate and rating performance.

Top films	Category	Rate and rating
Finding: Bucket Brotherhood is top with 34 rentals; some films rented only 4 times.	Finding: Sports $5,314, Sci-Fi $4,757, Animation $4,656 lead; Music $3,418 and Travel $3,550 are lowest.	Finding: $4.99 films bring $32,947 (49%), $2.99 bring $21,317 (32%), $0.99 bring $13,143 (19%). PG-13 earns most ($15,259), G least ($11,664).
Visual: top-N bar chart.	Visual: bar chart by category.	Visual: donut by rate, column by rating.

3.4 Inventory Analysis

Tables: inventory, film, rental

Copies per film	Films with no copy	Returns
Count copies of each film in each store.	Find films that cannot be rented.	Check whether copies are returned on time.
Finding: 4,581 copies of 1,000 films (store 1: 2,270, store 2: 2,311); about 3.5 rentals per copy.	Finding: 42 films have no copy.	Finding: 183 rentals never returned; about 46% of returned rentals (7,269 of 15,861) kept past the allowed days; average keep time is 5 days.
Visual: bar of copies per film or category.	Visual: table of films with no copy.	Visual: KPI cards and late vs on-time column.

3.5 Staff Analysis

Tables: staff, rental, payment

Rentals handled	Revenue	Average payment
Count rentals per staff member.	Compare revenue per staff member.	Compare payment size per staff member.
Finding: Mike 8,040 rentals, Jon 8,004.	Finding: Mike $33,482, Jon $33,924.	Finding: Mike $4.16, Jon $4.25. No customer rating data, so satisfaction cannot be measured.
Visual: clustered column.	Visual: clustered column.	Visual: card per staff member.

3.6 Stores Analysis

Tables: store, inventory, rental, customer

Store revenue	Busy days / hours	Customers per store
Compare revenue of the two stores.	Find the busiest weekdays and hours.	Compare customer counts and rental behavior.
Finding: store 1 $33,680, store 2 $33,727.	Finding: Tuesday busiest (2,463 rentals), Thursday quietest (2,200); busiest hour 3 PM (887).	Finding: store 1 has 326 customers vs 273, yet about 26.7 rentals each. About half of rentals happen at the non-home store.
Visual: column by store.	Visual: heatmap of weekday by hour.	Visual: card and column by store.

4. Why This Is MECE

- **Mutually exclusive:** Sales is money over time, Films is what is rented, Inventory is the copies, Staff is people, Stores is the two locations, Customers is who rents.
- **Collectively exhaustive:** customer behavior, sales trends, film performance, inventory, staff productivity and store operations are all covered.
- **Cross-area questions** such as revenue by category are handled in the combined analysis.

Area	Main tables
Customers	customer, address, city, country, payment
Sales	payment

Area	Main tables
Films	film, category, film_category, rental
Inventory	inventory, film, rental
Staff	staff, rental, payment
Stores	store, inventory, rental, customer

5. Combined Analysis

Combination	Insight
Films + Sales	$4.99 films rent as often as cheap ones, so they give much more revenue.
Films + Inventory	Top categories (Sports, Sci-Fi, Animation) are good places to add copies.
Customers + Stores	Customers rent about 26.7 films in both stores, so the store does not change how much they rent.
Inventory + Customers	Late and unreturned rentals keep copies away from other customers.

6. Recommendations

- **Marketing campaigns:** focus on India, China and the USA; reward top spenders; contact the 15 inactive customers; run offers on quiet days like Thursday.
- **Film collection:** add copies in Sports, Sci-Fi and Animation; keep $4.99 films; buy copies for the 42 films with none or remove them; review films with very few rentals.
- **Staff training and planning:** both staff perform equally, so train on following up late and unreturned rentals; add staff on Tuesdays, weekends and around 3 PM.
- **Store operations:** keep stock balanced across both stores because customers use both.

7. Power BI Dashboard Plan

Page	What it shows
Overview	Total revenue, total rentals, customers, average payment; monthly revenue; revenue by store
Customers	Customers by country, top 10 customers, active vs inactive
Films and Inventory	Top films, revenue by category, revenue by rental rate and rating, copies per film, films with no copy
Staff and Stores	Rentals and revenue per staff member, rentals by weekday and hour, store comparison

8. Limitations

- No customer rating or feedback data, so satisfaction cannot be measured directly.
- Data runs from May 2005 to February 2006 with most activity in July and August; read May, June and February trends with care.
- Only two stores and two staff members, so comparisons are limited.

9. Conclusion

Six non-overlapping areas, each with three sub-analyses, make the analysis clear. The store has loyal customers, balanced stores and equal staff performance. The best chances to improve are the film collection, inventory and returns, and staffing for busy times.